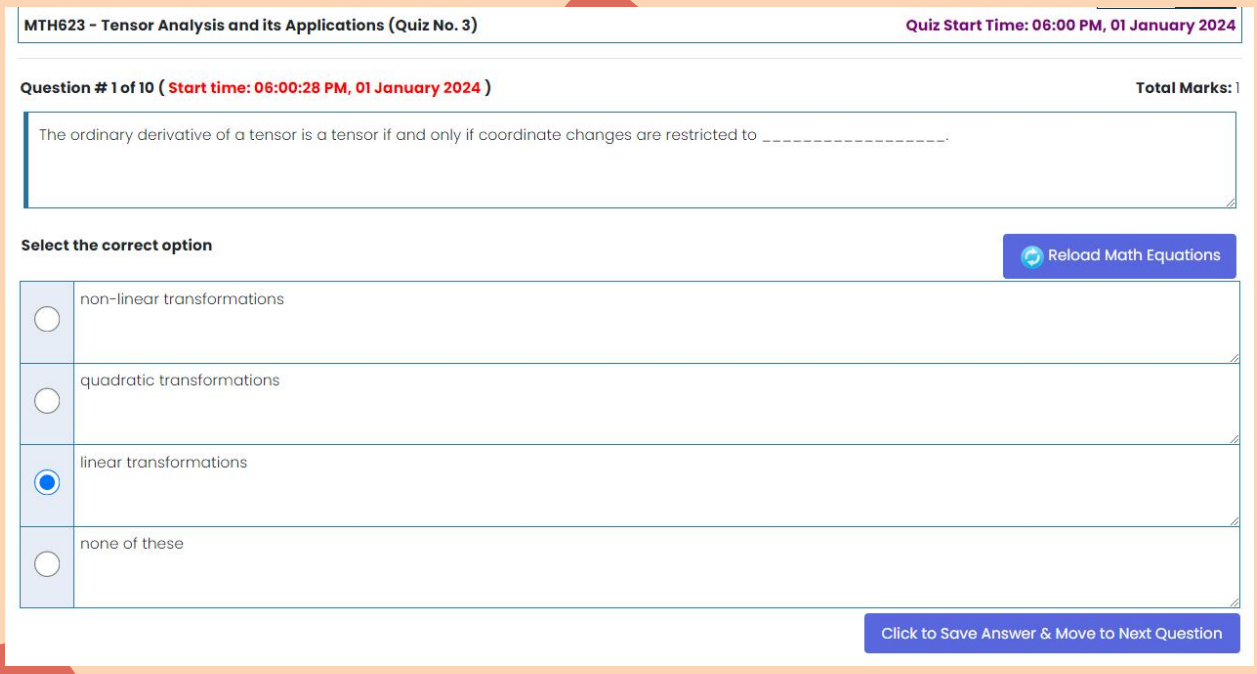
MTH 623

Quiz No.3

Solution by Sufyan

JUTT


MTH623 - Tensor Analysis and its Applications (Quiz No. 3)

Quiz Start Time: 06:00 PM, 01 January 2024

Question # 1 of 10 (Start time: 06:00:28 PM, 01 January 2024)

Total Marks: 1

The ordinary derivative of a tensor is a tensor if and only if coordinate changes are restricted to _____.

Select the correct option

Reload Math Equations

- non-linear transformations
- quadratic transformations
- linear transformations
- none of these

Click to Save Answer & Move to Next Question

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MTH623 - Tensor Analysis and its Applications (Quiz No. 3) — **Quiz Start Time: 06:00 PM, 01 January 2024**

Question # 2 of 10 (Start time: 06:01:02 PM, 01 January 2024) — **Total Marks:** 1

The ordinary derivative of a tensor is a tensor in general.

Select the correct option

Reload Math Equations

- ○ True
- ◉ False

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) — **Quiz Start Time: 06:00 PM, 01 January 2024**

Question # 3 of 10 (Start time: 06:01:29 PM, 01 January 2024) — **Total Marks:** 1

In generalized curvilinear coordinates the gradient of scalar function $\phi = \phi(x^i)$ is defined as _____.

Select the correct option

Reload Math Equations

- ○ $\nabla\phi = \frac{\partial\phi}{\partial x^i}$
- ◉ $\nabla\phi = \frac{\partial\phi}{\partial x^i}\hat{e}_i$
- ○ $\nabla\phi = \partial\phi\hat{e}_i$
- ○ $\nabla\phi = \frac{\partial x^i}{\partial\phi}$

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 06:00 PM, 01 January 2024

Question # 4 of 10 (Start time: 06:02:29 PM, 01 January 2024) **Total Marks:** 1

The covariant differentiation of a tensor with constant components is zero.

Select the correct option

Reload Math Equations

- (●) True
- ( ) False

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 06:00 PM, 01 January 2024

Question # 5 of 10 (Start time: 06:03:09 PM, 01 January 2024) **Total Marks:** 1

The Christoffel symbols of first kind Γ_{ijk} are symmetric in_____.

Select the correct option

Reload Math Equations

- ( ) first and third indices
- (●) first two indices
- ( ) all indices
- ( ) second and third indices

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 06:00 PM, 01 January 2024

Question # 6 of 10 (Start time: 06:04:26 PM, 01 January 2024) Total Marks: 1

Which of the following is true for basis vectors in general curvilinear coordinates systems?

Select the correct option

Reload Math Equations

- $\frac{\partial \vec{e}_j}{\partial x^i} = \Gamma^k_{ij} \vec{e}_k$
- $\frac{\partial \vec{e}_j}{\partial x^i} = \Gamma^k_{ij}$
- (●) $\frac{\partial \vec{e}_j}{\partial x^i} = 0$
- $\frac{\partial \vec{e}_j}{\partial x^i} = \vec{e}_k$

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 06:00 PM, 01 January 2024

Question # 7 of 10 (Start time: 06:05:19 PM, 01 January 2024) Total Marks: 1

The covariant differentiation of metric tensor is _____.

Select the correct option

Reload Math Equations

- zero in rectangular Cartesian coordinates only
- (●) zero in all coordinate systems

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 06:00 PM, 01 January 2024

Question # 8 of 10 (Start time: 06:06:00 PM, 01 January 2024) **Total Marks:** 1

In Cartesian coordinate system all the Christoffel symbols are _____.

Select the correct option

Reload Math Equations

- ○ vanish
- ○ all options are true
- ◉ zero
- ○ equal

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 06:00 PM, 01 January 2024

Question # 9 of 10 (Start time: 06:06:29 PM, 01 January 2024) **Total Marks:** 1

The covariant derivative of an arbitrary tensor is a tensor.

Select the correct option

Reload Math Equations

- ◉ True
- ○ False

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) **Quiz Start Time: 06:00 PM, 01 January 2024**

Question # 10 of 10 (Start time: 06:07:39 PM, 01 January 2024) **Total Marks:** 1

In curvilinear coordinates the derivative of a vector also includes_____.

Select the correct option

Reload Math Equations

- ○ the derivatives of unit vectors
- ◉ the derivatives of basis vectors

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) **Quiz Start Time: 12:28 PM, 01 January 2024**

Question # 2 of 10 (Start time: 12:29:26 PM, 01 January 2024) **Total Marks:** 1

The ordinary derivative of a tensor is a tensor if and only if coordinate changes are restricted to _____.

Select the correct option

Reload Math Equations

- ○ non-linear transformations
- ◉ linear transformations
- ○ quadratic transformations
- ○ none of these

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 12:28 PM, 01 January 2024

Question # 3 of 10 (Start time: 12:30:13 PM, 01 January 2024) Total Marks: 1

The Christoffel symbols of first kind Γ_{ijk} are symmetric in_____.

Select the correct option

Reload Math Equations

- all indices
- second and third indices
- first and third indices
- first two indices

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 12:28 PM, 01 January 2024

Question # 4 of 10 (Start time: 12:31:43 PM, 01 January 2024) Total Marks: 1

In curvilinear coordinates the derivative of a vector also includes_____.

Select the correct option

Reload Math Equations

- the derivatives of unit vectors
- the derivatives of basis vectors

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) **Quiz Start Time: 12:28 PM, 01 January 2024**

Question # 5 of 10 (Start time: 12:32:51 PM, 01 January 2024) **Total Marks:** 1

In Cartesian coordinate system all the Christoffel symbols are _____.

Select the correct option

Reload Math Equations

- ○ equal
- ○ all options are true
- ○ vanish
- ◉ zero

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) **Quiz Start Time: 12:28 PM, 01 January 2024**

Question # 6 of 10 (Start time: 12:33:35 PM, 01 January 2024) **Total Marks:** 1

The covariant differentiation of a tensor with constant components is zero.

Select the correct option

Reload Math Equations

- ◉ True
- ○ False

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 12:28 PM, 01 January 2024

Question # 7 of 10 (Start time: 12:34:13 PM, 01 January 2024) **Total Marks:** 1

In generalized curvilinear coordinates the gradient of scalar function $\phi = \phi(x^i)$ is defined as _____.

Select the correct option

Reload Math Equations

- ○ $\nabla\phi = \frac{\partial\phi}{\partial x^i}$
- ○ $\nabla\phi = \partial\phi\hat{e}_i$
- ○ $\nabla\phi = \frac{\partial x^i}{\partial\phi}$
- ◉ $\nabla\phi = \frac{\partial\phi}{\partial x^i}\hat{e}_i$

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 12:28 PM, 01 January 2024

Question # 8 of 10 (Start time: 12:35:41 PM, 01 January 2024) **Total Marks:** 1

The Christoffel symbols of second kind are symmetric in_____.

Select the correct option

Reload Math Equations

- ◉ lower indices
- ○ all indices

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) — **Quiz Start Time: 12:28 PM, 01 January 2024**

Question # 9 of 10 (Start time: 12:36:17 PM, 01 January 2024) — **Total Marks:** 1

In any coordinate system x^i, the covariant derivative with respect to x^k of a covariant vector T_i is defined as_____

Select the correct option

Reload Math Equations

- ○ $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^i_{ik} T_t$
- ◉ $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^t_{ik} T_t$
- ○ $T_{i;k} = \frac{\partial T_i}{\partial x^k} - \Gamma^t_{ik}$
- ○ $T_{i;k} = \frac{\partial T_i}{\partial x^k} + \Gamma^t_{ik} T_t$

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) — **Quiz Start Time: 02:04 PM, 30 December 2023**

Question # 1 of 10 (Start time: 02:04:35 PM, 30 December 2023) — **Total Marks:** 1

In the notation Γ^k_{ij} of Christoffel symbol of second, the symbol k tells_____

Select the correct option

Reload Math Equations

- ○ which basis vector points in the direction of this component of the derivative vector
- ○ which basis vector's change is being considered
- ○ none of these
- ◉ which coordinate is being varied to cause a change in the basis vector

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 02:04 PM, 30 December 2023

Question # 2 of 10 (Start time: 02:05:26 PM, 30 December 2023) **Total Marks:** 1

The Christoffel symbols of second kind are symmetric in_____.

Select the correct option

Reload Math Equations

- ○ all indices
- ◉ lower indices

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 02:04 PM, 30 December 2023

Question # 3 of 10 (Start time: 02:06:31 PM, 30 December 2023) **Total Marks:** 1

The ordinary derivative of a tensor is a tensor if and only if coordinate changes are restricted to _____.

Select the correct option

Reload Math Equations

- ○ none of these
- ◉ linear transformations
- ○ non-linear transformations
- ○ quadratic transformations

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 02:04 PM, 30 December 2023

Question # 5 of 10 (Start time: 02:08:44 PM, 30 December 2023) **Total Marks:** 1

In Cartesian coordinate system all the Christoffel symbols are _____.

Select the correct option

Reload Math Equations

- ○ vanish
- ○ equal
- ◉ zero
- ○ all options are true

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 02:04 PM, 30 December 2023

Question # 6 of 10 (Start time: 02:09:27 PM, 30 December 2023) **Total Marks:** 1

In generalized curvilinear coordinates the gradient of scalar function $\phi = \phi(x^i)$ is defined as _____.

Select the correct option

Reload Math Equations

- ○ $\nabla\phi = \frac{\partial\phi}{\partial x^i}$
- ○ $\nabla\phi = \frac{\partial x^i}{\partial\phi}$
- ○ $\nabla\phi = \partial\phi\hat{e}_i$
- ◉ $\nabla\phi = \frac{\partial\phi}{\partial x^i}\hat{e}_i$

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 02:04 PM, 30 December 2023

Question # 7 of 10 (Start time: 02:10:30 PM, 30 December 2023) Total Marks: 1

The Christoffel symbols of first kind Γ_{ijk} are defined by _____.

Select the correct option

Reload Math Equations

- $\Gamma_{ij} = \frac{1}{2}(g_{jk,i} + g_{ki,j} + g_{ki,j})$
- $\Gamma_{ijk} = \frac{1}{2}(-g_{jk,i} + g_{ki,j} + g_{ki,j})$
- $\Gamma_{jki} = \frac{1}{2}(-g_{jk,i} + g_{ki,j} + g_{ki,j})$
- $\Gamma_{ijk} = \frac{1}{2}(g_{jk,i} + g_{ki,j} + g_{ki,j})$

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 02:04 PM, 30 December 2023

Question # 8 of 10 (Start time: 02:11:59 PM, 30 December 2023) Total Marks: 1

The ordinary derivative of a tensor is a tensor in general.

Select the correct option

Reload Math Equations

- False
- True

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 02:04 PM, 30 December 2023

Question # 9 of 10 (Start time: 02:12:44 PM, 30 December 2023) Total Marks: 1

The covariant differentiation of a tensor with constant components is zero.

Select the correct option

Reload Math Equations

- (●) True
- ( ) False

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 02:04 PM, 30 December 2023

Question # 10 of 10 (Start time: 02:13:18 PM, 30 December 2023) Total Marks: 1

Christoffel symbols of first and second kind include derivatives of _____.

Select the correct option

Reload Math Equations

- (●) metric tensor
- ( ) normal vectors
- ( ) basis vectors
- ( ) unit vectors

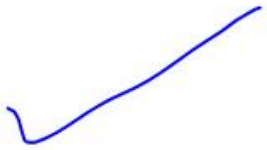
Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 12:51:20 PM, 01 January 2024)

The covariant derivative of an arbitrary tensor is a tensor.

Select the correct option

○	True
○	False



MTH623 - Tensor Analysis and its Applications (Quiz No. 3)

Question # 2 of 10 (Start time: 12:49:48 PM, 01 January 2024)

For a diagonal metric which of the following is true?

Select the correct option

- $\Gamma_{iii} = \frac{1}{2} g_{ii,i}$
- $\Gamma_{iii} = \frac{1}{2} g_{ij,j}$

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MTH623 - Tensor Analysis and its Applications (Quiz No. 3)

Question # 10 of 10 (Start time: 12:56:20 PM, 01 January 2024)

The Christoffel symbols of first kind satisfy the transformation law of a third rank tensor.

Select the correct option

- True
- False



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MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 02:57 PM, 01 January 2024

Question # 4 of 10 (Start time: 03:01:49 PM, 01 January 2024) Total Marks: 1

The covariant differentiation of metric tensor is _____.

Select the correct option

Reload Math Equations

- zero in all coordinate systems
- zero in rectangular Cartesian coordinates only

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz No. 3) Quiz Start Time: 03:20 PM, 01 January 2024

Question # 2 of 10 (Start time: 03:22:13 PM, 01 January 2024) Total Marks: 1

The covariant differentiation of a scalar is equal to _____.

Select the correct option

Reload Math Equations

- scalar field itself
- partial derivative of scalar field

Click to Save Answer & Move to Next Question